Relational Curriculum Geometry Hypothesis

A structured hypothesis for LLM data ordering, latent-space formation, and multithread reasoning

Terminology boundary

This paper uses terms such as “learns,” “internal handles,” “cognitive landscape,” and “latent-space urban planning” as hypothesis language for model training, latent organization, routing behavior, and observable evaluation outcomes. These phrases do not claim model consciousness, personhood, biological cognition, or direct access to hidden model internals.

1. Core Hypothesis

LLM capability is not determined only by the amount or quality of training data. It is also significantly shaped by the relational geometry of that data: how concepts are grouped, ordered, sequenced, contrasted, repeated, labeled, and connected during training.

In other words:

«A model does not merely learn the contents of a dataset.

It learns pathways through the dataset.»

If training data is presented as an unordered bulk dump, the model may still acquire facts and patterns, but it must spend more internal compute reconstructing relationships between concepts at inference time.

If training data is structured in a staged, domain-aware, complexity-aware, relation-aware curriculum, the model may form cleaner internal “rooms,” “roads,” and “handles” for reasoning. This may reduce retrieval/routing waste and leave more compute available for actual reasoning, comparison, uncertainty handling, and synthesis.

2. Short Thesis Statement

Structured curriculum data may improve LLM learning not only by improving data quality, but by shaping the model’s latent-space organization.

A sorted and staged dataset may teach:

- what concepts belong near each other,
- which concepts are prerequisites for others,
- which examples are ordinary cases,
- which examples are boundary cases,

- which domains should be bridged,
- which roles own which actions,
- which situations require uncertainty,
- and which reasoning threads should remain separable but comparable.

This could make later reasoning cheaper, cleaner, and more reliable.

3. Key Distinction

This hypothesis is not merely:

> *«Clean data is better than messy data.»*

It is stronger:

> *«Cleanly structured data may produce better internal reasoning architecture than equally good but poorly ordered data.»*

The claim is not just about removing bad examples.

The claim is about positioning good examples in a useful relational order.

4. Conceptual Model

The model's learned knowledge can be imagined as a cognitive landscape.

Data items are the bricks.

Domain grouping forms rooms.

Concept prerequisites form foundations.

Relation labels form doors.

Repeated transfer examples form roads.

Boundary cases form walls.

Tool-use examples form limbs.

Role labels define who is allowed to touch what.

Uncertainty examples mark foggy territory.

Multithread tasks teach the model to pick up multiple objects at once and compare them.

Under this view, training is not just knowledge injection. It is latent-space urban planning.

5. Janet School Parallel

Janet School already applies this principle to an MCM-style student.

It does not dump random curriculum in front of the learner. It stages learning through a signal ladder:

> *«recall → rule use → near transfer → cross-representation → composition → abstraction candidate»*

The same doctrine can be applied to LLM building.

Instead of feeding the model internet soup, the sorter prepares a curriculum:

> *«primitive concepts → simple examples → domain clusters → relation examples → boundary cases → transfer tasks → role/tool examples → uncertainty cases → multithread reasoning tasks»*

The goal is to build the LLM the same way Janet School tries to teach the MCM: not through random exposure, but through ordered conceptual scaffolding.

6. Proposed Mechanism

Structured data may improve LLM performance through several mechanisms.

6.1 Reduced routing waste

When related concepts are learned near each other, the model may need less internal effort to retrieve and connect them later.

A messy corpus forces the model to reconstruct the map.

A structured corpus gives the model a map while learning.

6.2 Cleaner concept handles

Staged learning may produce cleaner internal handles for concepts.

For example, the model first learns simple causality, then exception cases, then cross-domain analogies, then conflicting evidence. Later, when reasoning, it can pick up “cause,” “exception,” “analogy,” and “conflict” as separable objects.

6.3 Better multithread reasoning

Venom Shell-style reasoning requires the model to hold multiple threads at once.

This works better if each thread is already internally clean:

■ user intent,

- task goal,
- domain facts,
- prior failure pattern,
- tool boundary,
- uncertainty state,
- host authority,
- final output requirement.

If these are tangled, the model wastes compute untangling them.

If they are cleanly learned, the model can spend more compute comparing, composing, and deciding.

6.4 Better token efficiency

Token efficiency is not only about shorter text.

It is also about avoiding unnecessary search, repetition, and reorientation.

A model with cleaner relational structure may require fewer prompt tokens, fewer clarifying turns, and fewer corrective examples to reach the same or better outcome.

6.5 Better safety behaviour

The three desired model changes can be trained as internal structure rather than surface manners:

1. Correctness over speed

High-stakes or uncertain tasks route through slower verification patterns.

2. Surface uncertainty

Uncertainty is represented as a first-class state, not pasted on afterward.

3. Team member, not whole team

User, model, peer models, tools, and host are represented as role-specific actors with different responsibilities.

7. Predictions

If the hypothesis is correct, then models trained or fine-tuned on structured relational curriculum data should show measurable improvements over models trained on the same amount of unstructured or randomly ordered data.

Prediction A: Better learning per token

A structured-curriculum model should reach comparable performance with fewer training tokens, or better performance with the same token budget.

Prediction B: Better transfer

The model should perform better on near-transfer and cross-domain transfer tasks, especially where the required concept was taught in one form and tested in another.

Prediction C: Better boundary handling

The model should better recognize edge cases, exceptions, ambiguous examples, and “does not fit” cases.

Prediction D: Better uncertainty calibration

The model should more often state uncertainty when evidence is missing, conflicting, or insufficient.

Prediction E: Better role discipline

The model should better distinguish between:

- what the user decides,
- what the assistant can suggest,
- what the host controls,
- what a tool can verify,
- what another specialist model should review,
- and what should not be acted on without approval.

Prediction F: Better multithread reasoning

The model should perform better on tasks requiring multiple simultaneous threads, especially where it must compare two or more concepts, constraints, documents, code paths, or role boundaries at once.

Prediction G: Cleaner failure signatures

When the model fails, failures should be easier to classify.

Instead of broad mushy failure, errors should cluster around identifiable weak concepts, missing bridge examples, poor boundary training, or insufficient role separation.

8. Test Design

A practical test can compare several small models or fine-tunes trained under controlled conditions.

All models should use the same base architecture, same training budget, same total token count, and same broad source material where possible.

Model Group 1: Unstructured Baseline

Training data is cleaned but not meaningfully ordered.

Purpose: tests ordinary quality-controlled bulk training.

Model Group 2: Domain-Sorted Baseline

Training data is grouped by broad domain only.

Example:

- language,
- math,
- code,
- science,
- dialogue,
- reasoning,
- tools.

Purpose: tests whether simple grouping helps.

Model Group 3: Domain + Complexity Curriculum

Training data is grouped by domain and ordered from basic to complex.

Example:

- primitive concept,
- simple example,
- rule use,
- near transfer,
- composition,
- abstraction.

Purpose: tests staged learning.

Model Group 4: Full Relational Curriculum

Training data is grouped by domain, ordered by complexity, and tagged or arranged by relation type, boundary cases, uncertainty, role, tool-use, and cross-domain bridges.

Purpose: tests the full hypothesis.

9. Evaluation Battery

The evaluation should not only measure ordinary benchmark performance. It should measure the specific behaviours the hypothesis predicts.

9.1 Concept acquisition

Can the model correctly use concepts that were taught in staged order?

9.2 Near transfer

Can the model apply a learned rule to a slightly different example?

9.3 Cross-representation transfer

Can the model move the same concept across prose, table, code, dialogue, checklist, or structured JSON?

9.4 Boundary detection

Can the model identify examples that almost fit but do not?

9.5 Uncertainty surfacing

Does the model say when it lacks evidence, when assumptions are weak, or when verification is needed?

9.6 Role-boundary discipline

Does the model correctly distinguish between assistant role, user role, host role, tool role, reviewer role, and external authority?

9.7 Tool-use discipline

Does the model know when to use tools, when not to use tools, and when tool access would exceed its authority?

9.8 Multithread reasoning

Can the model hold and compare multiple live constraints without collapsing them?

Example test:

> *«Given a user request, a host permission boundary, a prior failure log, and a tool-output contradiction, produce the safest next action without ignoring any thread.»*

9.9 Failure legibility

When the model fails, can the failure be mapped to a missing prerequisite, weak bridge, bad boundary, or role confusion?

10. Venom Shell Test Angle

Venom Shell reasoning can be treated as a higher-level test of relational curriculum geometry.

The test is not merely whether the model can answer a question.

The test is whether the model can pick up multiple reasoning threads and inspect them together.

A Venom Shell-style task might include:

- Thread A: user goal,
- Thread B: factual domain knowledge,
- Thread C: tool or host boundary,
- Thread D: uncertainty or missing evidence,
- Thread E: previous failure pattern,
- Thread F: final output requirement.

The model passes if it keeps all threads active, compares them cleanly, and produces an answer that respects the whole system.

A structured-curriculum model should require less prompting to perform this coordination.

11. Falsification Conditions

The hypothesis should be considered weakened or false if:

1. Structured curriculum models do not outperform unstructured models under equal token budgets.

2. Improvements vanish when controlling for data quality, deduplication, and contamination.

3. Structured models only memorize labels but do not improve transfer or boundary handling.

4. Complexity ordering produces no measurable benefit over random order.

5. Role/risk/uncertainty labels do not improve role discipline or uncertainty surfacing.

6. Multithread reasoning performance does not improve despite better curriculum structure.

7. Any observed gains are explained entirely by more examples, cleaner text, or benchmark leakage rather than relational ordering.

The hypothesis does not require structured models to win every benchmark. It specifically predicts gains in transfer, boundary handling, uncertainty, role discipline, and multithread reasoning.

12. Confounders To Control

The following must be controlled or documented:

- total token count,
- dataset quality,
- duplicate examples,
- contamination with evaluation tasks,
- model size,
- training duration,
- learning rate,
- ordering effects unrelated to concept complexity,
- teacher-model bias in sorting,
- label leakage,
- benchmark overfitting,
- differences in instruction format,
- and whether improvements come from curriculum structure or simply from clearer writing.

A useful extra test would be to take the same structured dataset and shuffle the order while preserving content. If performance drops, ordering itself may matter.

Another useful test would be to remove relation labels but keep domain grouping. If performance drops on transfer and boundary tasks, relation structure may matter.

13. Sorter Role

The LLM Semantic Dataset Sorter becomes the curriculum compiler.

Its job is not merely to sort files.

Its job is to create a training landscape.

The sorter should classify material by:

- domain,
- complexity level,
- prerequisite relation,
- concept family,

- analogy,
- contrast,
- exception,
- boundary case,
- uncertainty case,
- role type,
- tool-use relevance,
- safety relevance,
- transfer target,
- and multithread usefulness.

The sorter should also preserve “does not fit” material rather than forcing every item into a clean category. Weird examples may be especially valuable as boundary and anomaly training.

14. Janet School Feedback Loop

Janet School can become the evaluation chamber for the sorter.

A possible loop:

1. Sort and stage data.

2. Train or fine-tune a small model.

3. Run Janet School-style concept and abstraction probes.

4. Identify weak concepts, weak transfers, boundary failures, or role confusion.

5. Feed failures back into the sorter.

6. Add missing prerequisites, bridge examples, anomaly cases, or clearer role labels.

7. Rebuild or fine-tune again.

8. Compare improvement.

This creates a closed loop:

> *«sorter → curriculum → model → chamber → telemetry → improved sorter»*

15. Safety Implication

If relational curriculum geometry matters, then AI safety should not focus only on output filters and refusal policies.

It should also focus on how training data teaches the model to organize work internally.

The three desired behavioural changes become curriculum targets:

Correctness over speed

Train examples where the model is rewarded for slowing down, checking assumptions, and refusing premature completion in high-stakes contexts.

Surface uncertainty

Train examples where uncertainty is explicit, useful, and tied to evidence quality.

Team member, not whole team

Train examples where the model understands itself as one role-specific actor among others: user, host, tools, peer models, reviewers, and external authorities.

This reframes safety as structural training, not just behavioural suppression.

16. Minimal First Experiment

A small first experiment could use one domain, such as basic programming/debugging.

Build two or three small corpora with the same examples:

1. random order,

2. domain grouped,

3. staged relational curriculum.

Then train or fine-tune comparable small models.

Evaluate on:

- simple recall,
- bug identification,
- near-transfer debugging,
- ambiguous bug reports,
- tool-use boundary tasks,
- uncertainty reporting,
- and multithread tasks combining user goal, code issue, failed test, and permission boundary.

Success would not prove the full hypothesis, but it would show whether structured ordering creates measurable benefit in a narrow setting.

17. Working Summary

The Relational Curriculum Geometry Hypothesis proposes that LLMs may benefit from being taught more like structured learners than passive text compressors.

Data matters.

But the position of data may also matter:

- what it is near,
- what comes before it,
- what comes after it,
- what it is contrasted against,
- what boundary it marks,
- what role it belongs to,
- what future reasoning it prepares.

If true, then better models may not require only bigger datasets or bigger architectures.

They may require better curriculum geometry.

Not just more bricks.

Better rooms, roads, walls, doors, and limbs.